

AuraSense v2: The Algorithm and the Alarms

Where we are, in plain words • updated 30 September 2026, after the design was frozen • Kevin Kim

In short. The program reads brain waves and gives a risk score every 5 seconds; alarm rules turn that score into warnings. Trained on other people and tested on someone new, it barely beats guessing, and without the time of day it's at chance. Once it has also learned from the person's own earlier seizures, the brain waves carry a real signal. The frozen design warns about 1 in 6 seizures with about 3.4 false alarms a day, 2 to 3 times better than random alarms, using brain waves alone. That's real, but not yet a device someone could rely on. It will now be tested once on 100 patients it has never seen.

1. How the algorithm works

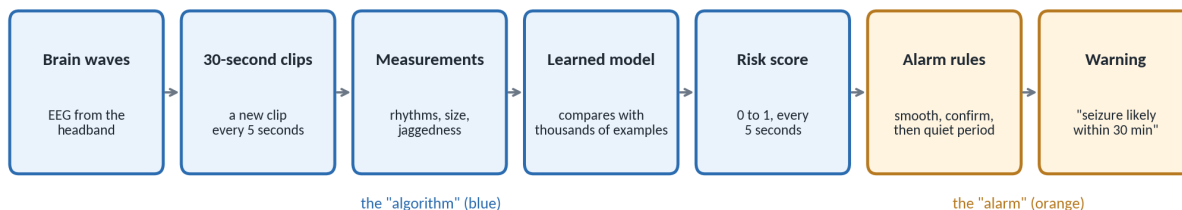


Figure 1. From brain waves to a warning. The blue steps are the "algorithm", which produces a risk score. The orange steps are the "alarm", which decides when to actually warn.

Every 5 seconds, the program takes the last 30 seconds of brain waves and measures simple things about them: how much slow and fast rhythm there is, how large the signal is, and how jagged it is. A learned model then compares those measurements with hundreds of thousands of examples it was trained on, where we knew whether a seizure followed within 30 minutes. It outputs a **risk score** between 0 and 1: its estimate of how much this moment looks like the half hour before a seizure.

2. How alarms work

A risk score alone isn't a warning. Raw scores flicker, so the alarm follows four rules:

- 1. Average.** The risk is averaged over the last 3 minutes, so one odd moment doesn't count.
- 2. Threshold.** An alarm needs the averaged risk to rise above a set level, the threshold.
- 3. Confirm.** It has to stay above the threshold for 30 seconds in a row.

4. Quiet period. After an alarm, the program stays quiet for 30 minutes, so one warning isn't repeated over and over.

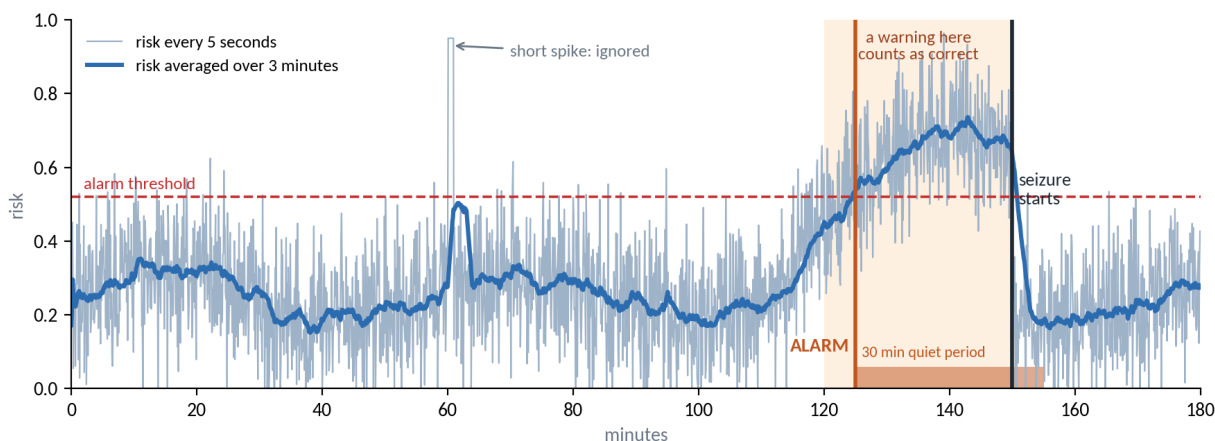


Figure 2. An illustration (made-up data) of one alarm. The pale line is the raw score, the dark line its 3-minute average. The short spike at minute 60 is ignored. The alarm fires when the average stays above the threshold, 25 minutes before the seizure, inside the orange stretch where a warning counts as correct.

A warning counts as correct if the seizure starts between 5 seconds and 30 minutes after it. Any other alarm is a **false alarm**. The threshold is the main dial. Set it low and more seizures are caught, but with more false alarms; set it high and there are fewer false alarms, but more seizures are missed. We choose it to keep false alarms at or below a target, such as 5 a day or 1 a day, using only data the program has already seen.

3. How accurate is the algorithm?

The score below asks: pick one moment from the half hour before a seizure and one normal moment. How often does the program rate the pre-seizure moment as riskier? 0.5 is a coin flip; 1.0 is perfect.

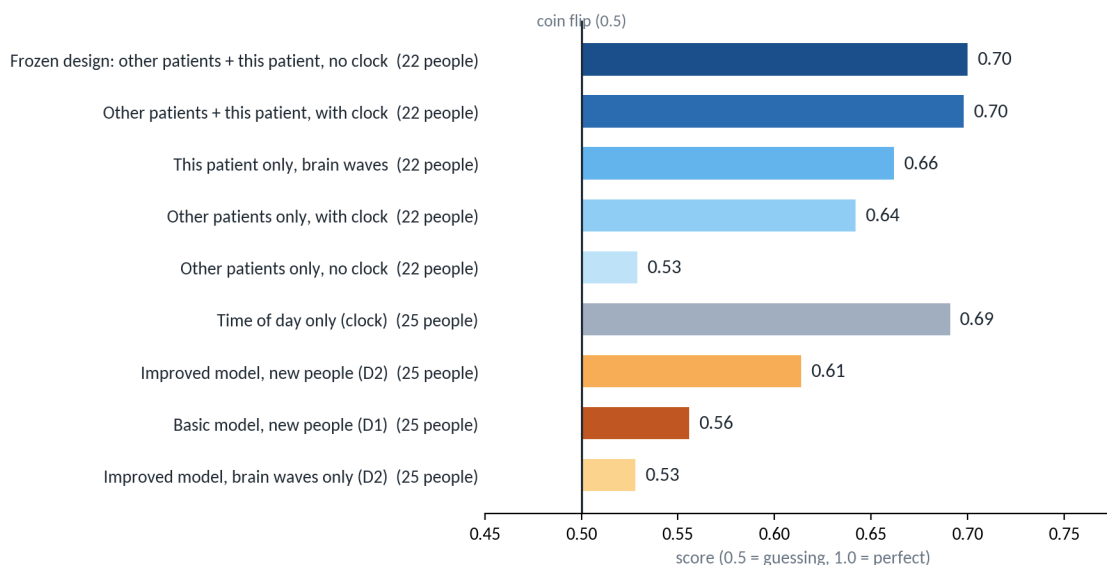


Figure 3. Accuracy of each version. Bars start at 0.5 (a coin flip). Blue bars are personalized designs, tested on 22 people with at least two seizures each. The others were tested on 25 people it had never seen. The grey "clock" bar is a program that knows only the time of day.

On people it has never seen, it barely works. The basic version (D1) scored 0.56. The improved version (D2) scored 0.61, but all of that gain came from knowing the time of day. Its brain-wave part alone scored 0.53, and a program that knows only the clock scored 0.69. In these hospital recordings, seizures clustered at certain times of day.

When it has seen the person before, it does better. Trained on the same person's other seizures, it scored 0.66 from brain waves alone. A model trained on everyone else and then also on this person's own data scored about 0.70, with or without the clock. Without the clock, the "everyone else" model alone drops to 0.53, a coin flip. So the warning signs in the brain waves seem to be real but personal: they differ from one person to the next. The frozen design (0.70) is the best version without the clock; that score is slightly optimistic, because it was picked on these same people.

4. How good are the alarms?

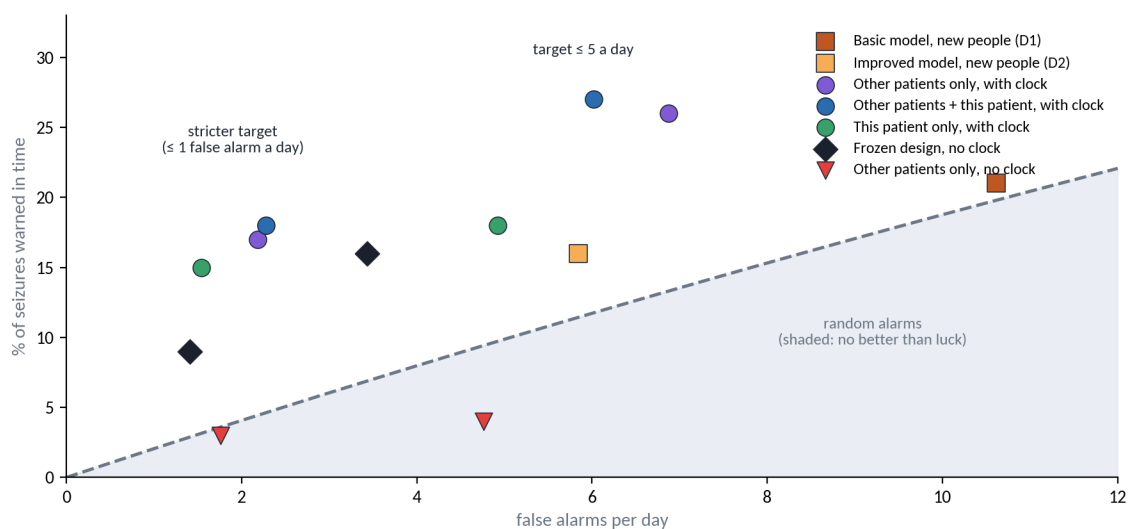


Figure 4. What a user would experience. Each point is one design: the share of seizures warned in time (higher is better) against false alarms per day (further left is better). Points in the shaded area would be no better than an alarm that goes off at random as often.

Trained on others and tested on someone new, the alarms are no better than random. D1 warned about 21% of seizures with 10.6 false alarms a day, and random alarms at that rate would catch about 20%. D2 warned 16% with 5.8 a day, which wasn't reliably better than luck either.

Personalized, the alarms are clearly better than random. Each person's threshold was chosen from their own earlier normal brain activity, never from the seizure being tested. With the clock, the best design warned about 27% of seizures at about 6 false alarms a day, where random alarms would catch 12%. The **frozen design, which uses no clock**, warned **16% at 3.4 false alarms a day** (random: 7%) and **9% at 1.4 a day** (random: 3%). These differences are statistically solid, not luck.

A correction to the first version of this note. It said that tuning the alarm threshold to a person's normal brain activity was enough to make a model trained on other people beat random alarms. That test included the time of day. Without the clock, the same model stays at or below random (4% warned at 4.8 false alarms a day), even with a personal threshold. For brain waves alone, **learning from the person's own seizures** is what makes the alarms beat random.

Design	False-alarm target	Seizures warned	False alarms a day	Random alarms would warn
Frozen design, no clock	≤ 5 a day	16%	3.4	7%
Frozen design, no clock	≤ 1 a day	9%	1.4	3%
Other patients only, no clock	≤ 5 a day	4%	4.8	9%

Design	False-alarm target	Seizures warned	False alarms a day	Random alarms would warn
Other patients + this patient, with clock	≤ 5 a day	27%	6.0	12%
Other patients only, with clock	≤ 5 a day	26%	6.9	13%
This patient only, with clock	≤ 5 a day	18%	4.9	10%
Other patients + this patient, with clock	≤ 1 a day	18%	2.3	5%
Other patients only, with clock	≤ 1 a day	17%	2.2	4%
This patient only, with clock	≤ 1 a day	15%	1.5	3%

Two honest limits. First, the numbers are still modest: about 1 in 6 seizures warned at 3.4 false alarms a day, or 1 in 11 at 1.4 a day, is far from a device someone could rely on. The course goal was 80% of seizures at no more than 5 false alarms a day. Second, false alarms can come out above the target (with the clock, about 6 instead of 5; for the frozen design, 1.4 instead of 1), because a person's brain activity drifts between the time the threshold is set and the time it's used. A threshold that keeps adjusting could help.

5. What happens next

The design is frozen. After a final round of pre-planned comparisons, the design was fixed: a model trained on other patients plus the person's own data, with brain-wave measurements and 10-minute trends, each person's normal levels as a baseline, and no clock. It is locked with a Git tag, so it can't be adjusted after anyone sees the final test.

Before the final test: a planned check of whether the results depend on the 4-hour definition of normal EEG (rerun with 1 hour), and an adapter for the behind-the-ear sensors in the SeizelT2 dataset, built and checked on 25 of its patients.

The final exam: the frozen design, run once on the other 100 SeizelT2 patients, which nobody has looked at. That gives the most trustworthy answer to whether the approach works on new people and a wearable-style sensor. After that come the write-up and D3: a screen that replays recordings with the risk score and alarms, and a live test with the OpenBCI headset.